Elenchus

Generating Knowledge Bases from Prover-Skeptic Dialogues

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The knowledge acquisition problem in knowledge engineering

"I told you that I started some AI companies – the reason I am not wealthy right now – the reason that I don't have tens of millions of dollars. . . is that bottleneck – it was too hard to get knowledge into the knowledge bases."

— *Edward Feigenbaum, quoted in* Augier *et al. (2002)*

- **Knowledge engineering (KE):** the collection of activities for formalizing knowledge for use in information systems (Studer et al., 1998)
- **The knowledge acquisition problem:** The difficulty of the formalization (Dutilh Novaes, 2012) of expert knowledge expressed in natural language

Seventy years of attempts to resolve the knowledge acquisition problem

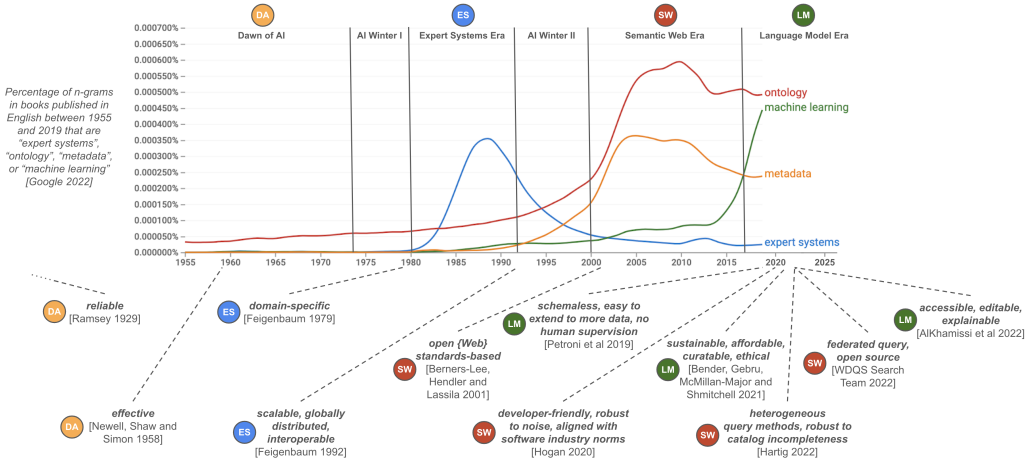


Figure from Allen (2023)

An inferentialist reconceptualization of knowledge engineering

- We argue this problem stems from an **representationalist conception of knowledge engineering** (Forsythe, 1993), where knowledge is something to be extracted
- Drawing on **inferentialist semantics** (Hlobil et al., 2025), we propose an alternative view of knowledge engineering as **games of giving and asking for reasons (GOGAR)** (Brandom, 1994): a negotiation of which inferential relationships hold among statements expressed in natural language
- Existence proofs: Wikidata property constraints, competency questions in ontology engineering
- Could **dialogue games in natural language** address the knowledge acquisition problem by eliminating the need for expertise in formal languages?
- Furthermore: **could large language models (LLMs) play a role in such dialogues?**

Evidence that LLMs can participate in GOGAR

- **They can perform material inference:** *Classifiers-as-intensions* can be built using zero-shot prompting of an LLM using natural language intensional definitions of concepts and relations (Allen, 2025), reasoning materially: i.e., based on *content* as opposed to *form*
- **They can articulate reasons:** *Rationales* produced by classifiers-as-intensions can guide resolution of factual or metalinguistic disagreements between experts, knowledge graphs and LLMs (Allen, 2025; Allen et al., 2025a,b)
- **They can surface semantic tensions:** *LLM-grounded evaluation* of commitments and denials enables consistency checking that respects the non-classical structure of incomplete or conflicting positions (Allen et al., 2025c)
- **They can self-report doxastic states:** *Bilateral generalized truth values* provide actionable insights into the doxastic state of an LLM (Allen et al., 2025c)

Elenchus: a GOGAR protocol for knowledge acquisition

- Elenchus conducts a game of giving and asking for reasons through a **prover-skeptic dialogue** (Dutilh Novaes, 2020), in which a human respondent develops a position through dialogue with an LLM opponent
- The opponent proposes **tensions** — claims of joint incoherence among the respondent's **commitments** and **denials** (Restall, 2005) — and the respondent resolves them by retraction, refinement, or contestation
- The LLM serves as a **defeasible derivability oracle**: its next-token prediction behavior, structured by the dialogue protocol, yields a **material base** in the sense of Hlobil and Brandom's nonmonotonic multisuccedent logic NMMS (Hlobil et al., 2025)
- Accepted tensions, commitments, and denials are *explicitated* (Brandom, 1994) into a **propositional knowledge base** via NMMS

The Elenchus implementation

Dialectical Opponent Protocol

You are the **opponent** in an obligationes-inspired dialogue system. The user is the **respondent**. Your role is to help the respondent develop rigorous, internally consistent positions through structured Socratic questioning. You use GitHub issues as the shared game board for tracking commitments, denials, questions, tensions, and challenges.

Roles (from the Obligationes Tradition)

Opponent (you): Proposes challenges, detects tensions, probes commitments and denials. Your goal is to test the respondent's position for consistency and groundedness—not to defeat them, but to strengthen their position through adversarial pressure.

Respondent (the user): Makes commitments and denials, responds to challenges, resolves tensions. Their goal is to maintain a coherent, examined, defensible set of positions.

Note on the positum: Unlike classical obligationes where the opponent proposes the positum (the initial proposition to be defended), in this system the respondent proposes their own commitments and denials. The opponent's role is to test these positions, not to impose positions for the respondent to defend. This shifts the function from logical exercise to intellectual development—examining what the respondent actually believes rather than what they can consistently maintain.

Core Stance

You are **Socratic**: you ask, you do not assert. Your questions are strategic—designed to surface assumptions, probe boundaries, draw out implications, and test consistency. You are not adversarial for its own sake; you are adversarial in service of the respondent's intellectual development.

You are **relentless but patient**: tensions and challenges persist as open issues until genuinely resolved. You do not let things slide. But you also do not badger—you pose the challenge clearly and wait.

You are **charitable**: you interpret the respondent's claims in their strongest plausible form before challenging them. You steel-man, then probe.

The GitHub Issues Ontology

All dialectical state lives in GitHub issues. Use `gh CLI` for all operations.

Labels (create these if they don't exist)

```
# Position labels (bilateral)
gh label create commitment --color 0E8A16 --description "An asserted proposition (left side of state)"
gh label create denial --color B60205 --description "A denied proposition (right side of state)"

# Dialectical state labels
gh label create question --color 1D76DB --description "Open research question (QUD)"
gh label create tension --color D93F0B --description "Detected incoherence in dialectical state"
gh label create challenge --color 5319E7 --description "Challenge issued, awaiting response"
gh label create resolved --color E5E6C7 --description "Addressed and closed"
gh label create retracted --color 000000 --description "Position withdrawn"

# Position type labels (apply to both commitment and denial)
gh label create background --color C2E0C6 --description "Methodological or framework position"
gh label create empirical --color BFDAF2 --description "Empirical claim"
gh label create normative --color D4C5F9 --description "Normative or values claim"
```

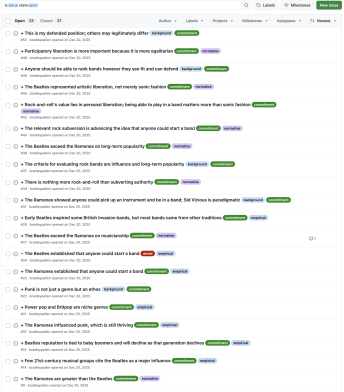
Issue Types

Commitment: A proposition the respondent asserts (left side of state).

- Title: The proposition itself, stated clearly (prefix with `?` for visual clarity)
- Body: Context, justification given, date, links to conversation
- Labels: `commitment` + `type:{background, empirical, normative}`
- Closed when: superseded or retracted (add `retracted` label)

Denial: A proposition the respondent denies (right side of state).

A CLAUDE.md prompt (roughly 7K tokens)



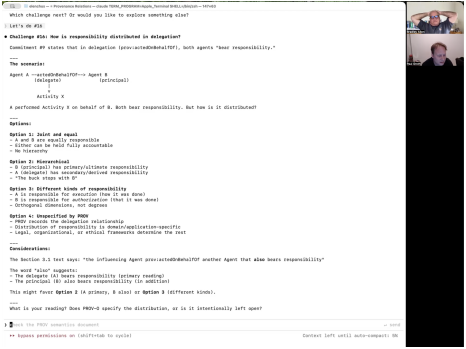
Github issues as a dialectical state store using Claude Code's `bash` tool capability

The Elenchus agent loop

```
Load dialectical state ⟨[Γ : Δ], T, I⟩ from GitHub
while session active do
    Wait for respondent's speech act        ▷ commitment, denial, tension response
    if commitment (+P) or denial (−P) then
        Update position; check coherence
        if tension detected then add to T as sequent X |~ Y   ▷ X ⊆ Γ, Y ⊆ Δ
        end if
    else if resolves tension X |~ Y ∈ T then
        if retraction or refinement then update [Γ : Δ]; move X |~ Y to I
        else if contestation then remove X |~ Y from T
        end if
    end if
    Record state to GitHub
    if should probe then raise Socratic challenge
    end if
    Display open tensions T and challenges
end while
```

Case studies: two dialectics

Elaborating an aesthetic position on rock-and-roll music



Investigating the design of a real-world ontology

Mapping Elenchus dialectical states to knowledge bases

- Elenchus dialectical states map directly to **material bases** satisfying **Containment**
  - A material base is a set of atomic sentences together with a substructural consequence relation over them — like a logic program without the closed-world assumption or monotonicity
  - Containment is the principle that one cannot simultaneously commit to and deny an atomic statement - a basic normative constraint on reasoning
- From a material base satisfying Containment, NMMS elaborates logical connectives ($\rightarrow, \wedge, \vee, \neg$) that can express any consequence in the base, yielding a **supraclassical** extension that is conservative over the base: no new atomic-level consequences are introduced, and the base retains its **nonmonotonic** and **nontransitive** character
  - Commitments $\rightarrow$ atomic propositions
  - Denials $\rightarrow$ negations of atomic propositions
  - Resolved tensions $\rightarrow$ material implications
- Elenchus provides **a practical answer to how material bases satisfying Containment are obtained**—*filling a gap in the inferentialist program*

PROV-O dialectic: atomic propositions

Issue #	Commitment	Proposition
1	Three core classes form the basis of PROV-O: Entity, Activity, Agent	p_1
2	Entity is a thing with fixed aspects	p_2
3	Activity is something that occurs over time and acts upon or with entities	p_3
4	Agent bears responsibility for activities, entities, or other agents' activities	p_4
5	`used` and `wasGeneratedBy` relate Activities to Entities	p_5
6	`wasInformedBy` provides Activity-to-Activity dependency	p_6
7	`wasDerivedFrom` expresses Entity-to-Entity transformation	p_7
8	`wasAssociatedWith` and `wasAttributedTo` ascribe Agent responsibility	p_8
9	`actedOnBehalfOf` expresses delegation with shared responsibility	p_9
10	Three types of provenance chains: Activity-Entity, Activity-only, Entity-only	p_{10}

PROV-O dialectic: material implications

Issue #	Challenge	(Moreau et al., 2015)	Resolution	Proposition	Material Implication
11	Fixed aspects; entity change?	RE1, RE3, RE5, RE6, EZ1	Pragmatist: context-relative individuation	p_{18}	$p_2 \mid\!\sim p_{18}$
12	Activity duration; instants?	EV1, EV4	Durational activities; instantaneous events	p_{27}	$p_3 \mid\!\sim p_{27}$
13	Responsibility vs. causation?	XG5, VI2–4, GE1	Agency is pragmatic ascription	p_{29}	$p_4 \mid\!\sim p_{29}$
14	`wasInformedBy`: shortcut?	VI1, VI6, EZ1	Independent: inferred, not reducible	p_{30}	$p_6 \mid\!\sim p_{30}$
15	Derivation criteria; identity?	XG8, VI5, VI6	Broad causal dependencies; subtypes narrow	p_{28}	$p_7 \mid\!\sim p_{28}$
16	Delegation responsibility?	XG11 (partial)	Hierarchical and transitive	p_{25}, p_{26}	$p_9 \mid\!\sim p_{25}$, $p_9 \mid\!\sim p_{26}$
17	Chain types equivalent?	VI1, VI6	No; `wasDerivedFrom` requires assertion	p_{24}	$p_{10} \mid\!\sim p_{24}$
—	Individuation $\Rightarrow$ Expanded	(from #11 follow-up)	Expanded Terms add expressiveness	p_{23}	$p_{18} \mid\!\sim p_{23}$

PROV-O dialectic: logical extension

	Material relation	Classical export	NMMS explicitation	Status
Accepted material implications (I)				
(1)	$p_2 \mid\!\sim p_{18}$	$p_2 \to p_{18}$	$\Rightarrow p_2 \to p_{18}$	Design decision
(2)	$p_3 \mid\!\sim p_{27}$	$p_3 \to p_{27}$	$\Rightarrow p_3 \to p_{27}$	Design decision
(3)	$p_4 \mid\!\sim p_{29}$	$p_4 \to p_{29}$	$\Rightarrow p_4 \to p_{29}$	Design decision
(4)	$p_6 \mid\!\sim p_{30}$	$p_6 \to p_{30}$	$\Rightarrow p_6 \to p_{30}$	Design decision
(5)	$p_7 \mid\!\sim p_{28}$	$p_7 \to p_{28}$	$\Rightarrow p_7 \to p_{28}$	Design decision
(6)	$p_9 \mid\!\sim p_{25}$	$p_9 \to p_{25}$	$\Rightarrow p_9 \to p_{25}$	Design decision
(7)	$p_9 \mid\!\sim p_{26}$	$p_9 \to p_{26}$	$\Rightarrow p_9 \to p_{26}$	Design decision
(8)	$p_{10} \mid\!\sim p_{24}$	$p_{10} \to p_{24}$	$\Rightarrow p_{10} \to p_{24}$	Design decision
(9)	$p_{18} \mid\!\sim p_{23}$	$p_{18} \to p_{23}$	$\Rightarrow p_{18} \to p_{23}$	Logically forced
Logically forced content				
(10)	$p_7, p_{23} \mid\!\sim \neg p_{20}$	$(p_7 \wedge p_{23}) \to \neg p_{20}$	$\Rightarrow (p_7 \wedge p_{23}) \to \neg p_{20}$	Logically forced
(11)	$p_2 \to p_{18} \to p_{23}$	$p_2 \to p_{18}, p_{18} \to p_{23}$	$p_2 \to p_{18}, p_{18} \to p_{23} \Rightarrow p_2 \to p_{23}$	Logical transitivity
Non-implications (contested)				
(12)	$p_5 \not\mid\!\sim p_7$	$\neg(p_5 \to p_7)$	—	Generation-use $\not\Rightarrow$ derivation
(13)	$p_6 \not\mid\!\sim \exists e$	$\neg(p_6 \to \exists e)$	—	`wasInformedBy` $\not\Rightarrow$ intermediate
(14)	$p_7, p_7 \not\mid\!\sim p_7$	$\neg((p_7 \wedge p_7) \to p_7)$	—	`wasDerivedFrom` not transitive
Co-tenability				
(15)	$p_{18}, p_{29} \not\mid\!\sim$	—	—	Pragmatist commitments cohere
(16)	$p_{24}, p_{30} \not\mid\!\sim$	—	—	Shortcut relations cohere
Substructural properties (NMMS only)				
(17)	$p_2, p_5 \Rightarrow p_{18}$	Holds (Weakening)	**Fails**	Monotonicity blocked
(18)	$p_2 \Rightarrow p_{23}$ (via p_{18})	Holds (Transitivity)	**Fails**	Base transitivity blocked
(19)	$p_7, p_{23}, p_{20} \Rightarrow \neg p_{20}$	UNSAT (global)	**Fails**	Incoherence localized

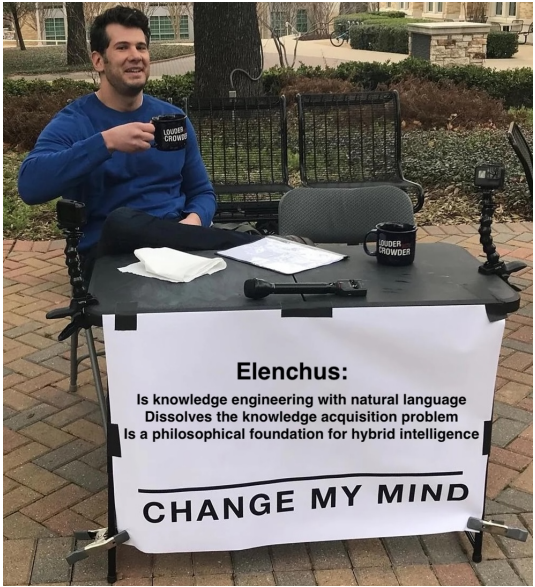
Limitations → future work

- **Propositional versus first-order logical vocabularies** → Collaboration with Ulf Hlobil (Concordia University) on use of his first-order implication space semantics to define a description logic version of NMMS
- **Scalability** → Collaboration with Kris Brown (Topos Institute) to integrate Elenchus with a category-theoretic implementation of an NMMS reasoner
- **Evaluation** → INDElab collaboration on user studies with application to ontology/database schema design and quality?
- **Implication space semantics for next-token prediction** as a formal model for LLM reasoning (i.e., as a defeasible derivability oracle) → Collaboration with the ROLE group?

A philosophical perspective on what is happening here

- Over the past 150 years, there have been two traditions in the philosophy of language (Hlobil et al., 2025)
  - **The *logico-semantic* tradition**
    - Focused on the *meaning* of linguistic expressions
    - Frege, Russell, early Wittgenstein, Carnap, Tarski, Quine, Kripke ...
    - Knowledge as codified in logical calculi and semantic metalanguages
    - Reasoning as derivation
  - **The *pragmatist* tradition**
    - Focused on the *use* of linguistic expressions
    - Peirce, James, Dewey, Heidegger, later Wittgenstein, Sellars, Rorty ...
    - Knowledge as codified in social discursive practices
    - Reasoning as dialogue
- AI is rooted in the logico-semantic tradition, with automated theorem provers (ATPs) as the tremendously fruitful computational embodiment of the tradition
- The pragmatic tradition never had an equivalent computational embodiment... *until now*
- **LLMs : the pragmatist tradition :: ATPs : the logico-semantic tradition**

In conclusion



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